

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457015.90055 +/- 0.00076 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1422 +/- 0.0083
Transit depth (Rp/Rs)^2	2.02 +/- 0.24 [%]
Orbital Inclination (inc)	83.17 +/- 1.43
Airmass coefficient 1 (a1)	0.9717 +/- 0.0067
Airmass coefficient 2 (a2)	0.0177 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	#3 - [84, 273]
Optimal Aperture	4.36
Optimal Annulus	20.16
Transit Duration (day)	0.0523 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1422 ± 0.0083 vs 0.1594 (deviation 1.94σ)
Duration fit vs published	0.0523 d vs 0.0512 d (deviation 0.25σ)
Mid-time O–C	1.8 min
Depth SNR	16.1
Residual scatter	0.69 %

Light curve

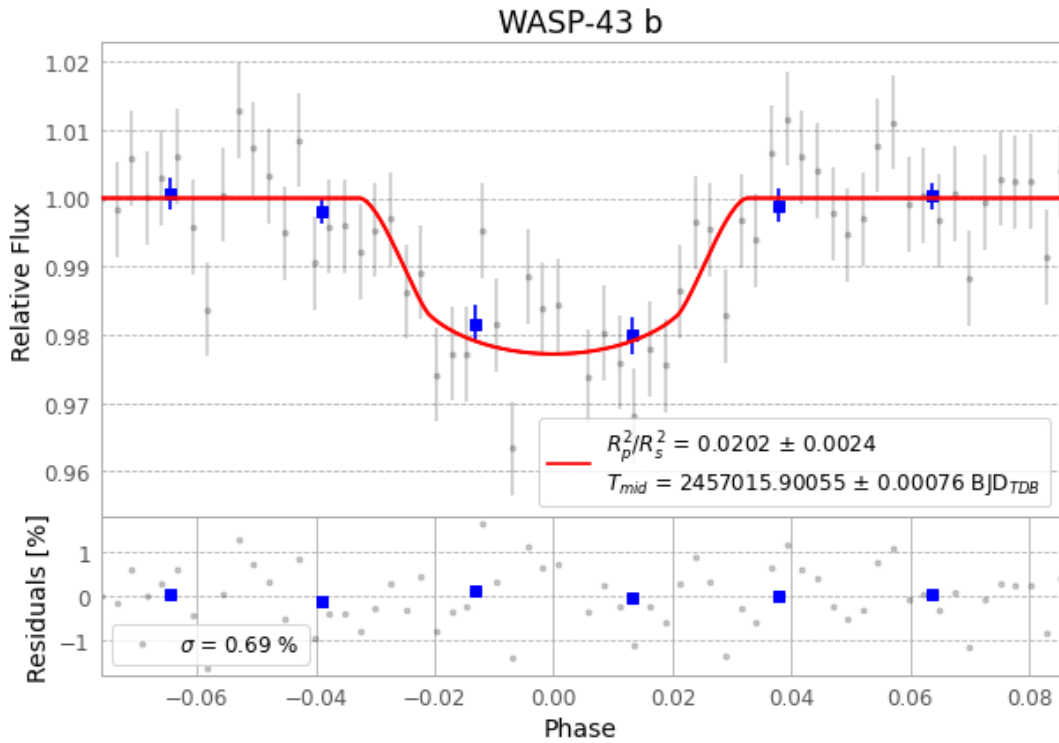


Figure 1 — FinalLightCurve_WASP-43 b_2014-12-24.png (SHA-256 b327819bad44f406...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [150, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ea8f5a5813ea498c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43141224080312.FITS → WASP-43141224111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-141224.log (bb6aa5f57e96...), FinalLightCurve_WASP-43 b_2014-12-24.png (b327819bad44...), FinalLightCurve_WASP-43 b_2014-12-24.csv (8d90127894b8...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:59Z.